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PART I

The Lifecycle at a Glance

1

The whole project on a few pages

1.1 The question

Public cervical cytology benchmarks are released as cell crops, and most crops are cut from far fewer microscope fields or slide images. When a benchmark is split at the crop level, crops from one field land on both sides of the train–test boundary. Files are disjoint, but the test set shares stain, illumination, focus, background, and sometimes literal pixels with training. The project asked three questions in order: *is the structure there*, *is the shared parent recognisable*, and *does it inflate reported performance*; and then a fourth: *how do models use it*.

1.2 What happened, in order

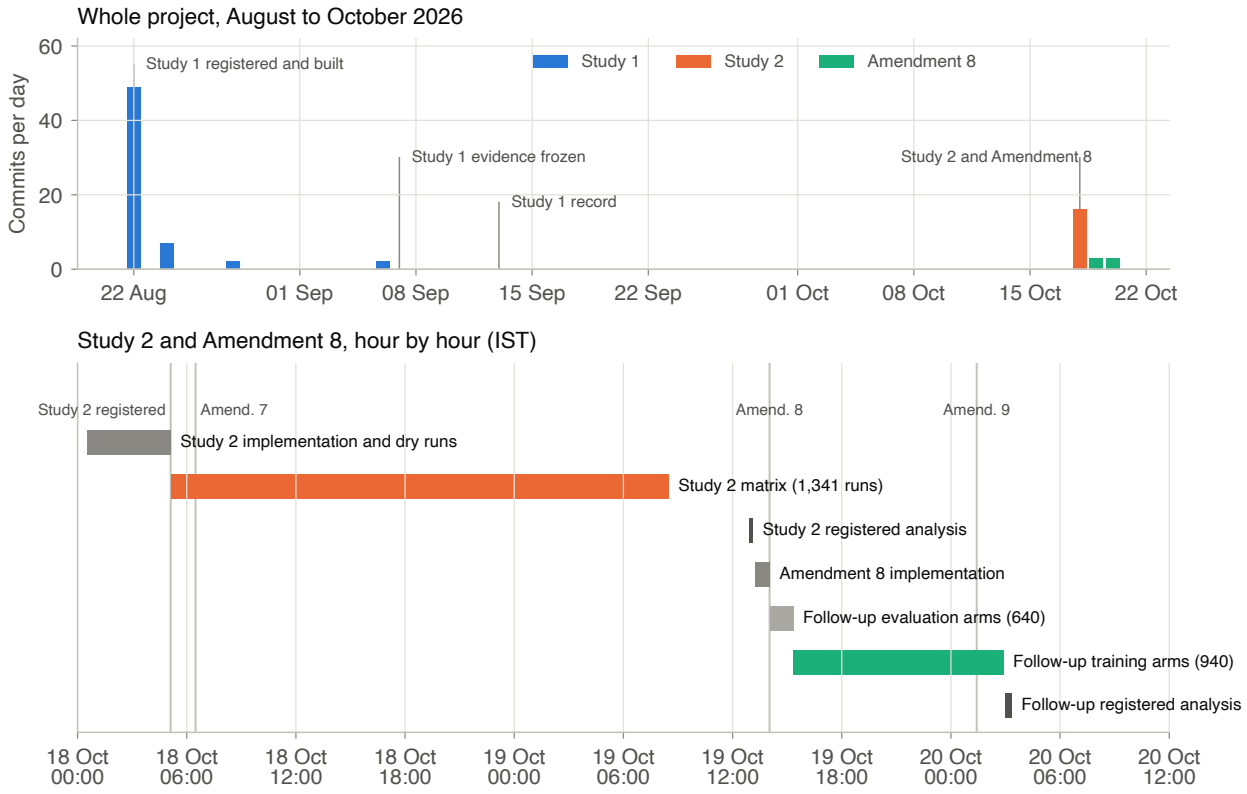


Figure 1.1: The lifecycle. Top: commits per day across the project. Bottom: Study 2 and Amendment 8 hour by hour; every span is measured from run records or commits.

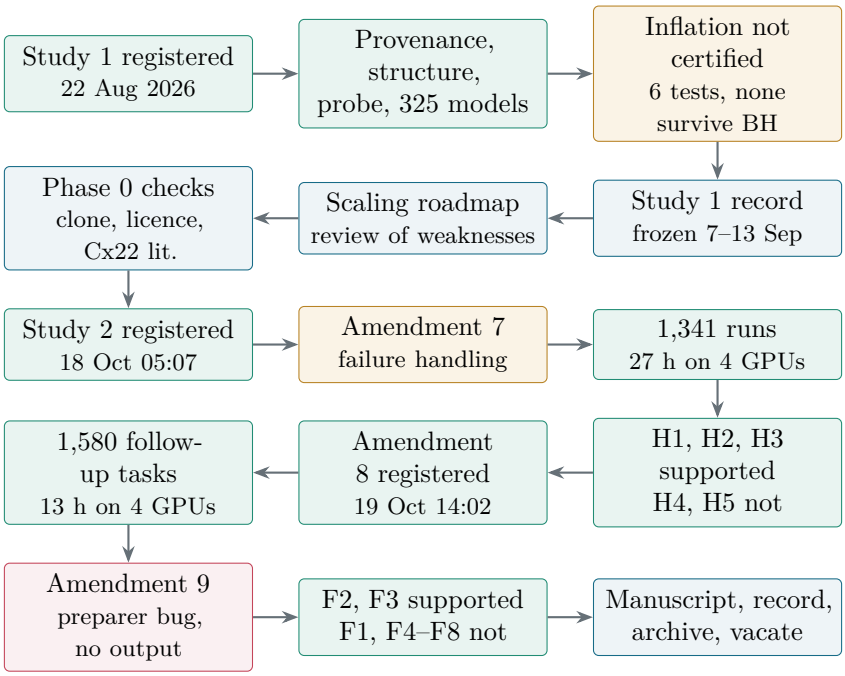


Figure 1.2: The whole lifecycle as a sequence of registered decisions (green), amendments and setbacks (gold, red), and reviews (blue). Every arrow is a commit in the repository; Appendix O lists all of them.

1. **Study 1 (22 August – 13 September 2026).** The crop-to-parent hierarchy of four public datasets was recovered with independent, fail-closed branches; random crop splitting placed a training parent behind 92.6% of SIPaKMeD and 99.97% of CRIC test cells; more than half of SIPaKMeD test crops shared source pixels with a training crop; a 424-way parent probe reached 109 times chance. Classifiers scored higher under crop-level splits for all six architectures and all seven Cx22 segmentation models, but none of the per-architecture tests survived the registered Benjamini–Hochberg correction. Part II is that record.

2. **The roadmap (October 2026).** A review identified four weaknesses: one question asked six times, availability not distinguished from use, leakage confounded with training diversity, and only one rescored benchmark. It proposed a single mixed-model test sized by simulation, causal arms, more architectures and seeds, preregistration, and keeping every per-sample artifact (Chapter 21).

3. **Study 2 (18–19 October).** Registered at commit `a62e8e7` (2026-10-18 05:07 IST), before any Study 2 model was trained: one primary test (H1) and a Holm family of four mechanism tests (H2–H5), 1,341 runs over 27.4 hours on four GPUs. H1, H2, and H3 were supported; H4 and H5 were not.

4. **Amendment 8 (19–20 October).** Eight new hypotheses about *how* the leak is used, registered at `80171fe` (2026-10-19 14:02 IST) after the Study 2 analysis and before any follow-up run: 1,580 tasks over 12.8 hours. F2 and F3 were supported; F1 went the opposite way; F4–F8 were null.

5. **Amendment 9 (20 October, 01:25).** A labelling bug in the follow-up preparer halted the analysis before it produced output; fixed and recorded before any affected estimate was seen.

6. **Close-out.** Manuscript (`paper/v2`), this record, 121 GB of outputs archived and verified on the network home, and the node-local disk vacated.

1.3 Every registered test

Table 1.1: Every registered confirmatory test of Study 2 and Amendment 8. H1 is tested alone; H2–H5 and F1–F8 are separate Holm families. Units as in the text (pp; slopes in pp per pp).

	Question	Estimate	95% CI	p	Holm p	Verdict
H1	SIPaKMeD P1 − P2 accuracy (primary)	0.97	0.59 to 1.39	<0.001	alone	**supported**
H2	CRIC P1 − P2 accuracy	12.08	11.16 to 12.99	<0.001	<0.001	**supported**
H3	2C leakage at matched diversity	1.73	0.95 to 2.51	0.006	0.02	**supported**
H4	2B own-context route	0.42	−0.35 to 1.18	0.23	0.23	**not supported**
H5	2A gap vs recoverability	0.059	−0.007 to 0.125	0.08	0.16	**not supported**
F1	CRIC slide-context pull	−4.29	−8.60 to −1.37	<0.001	0.001	**not supported**
F2	test-time stain transfer	1.27	0.38 to 2.16	0.005	0.03	**supported**
F3	leak with backgrounds masked	1.86	1.42 to 2.30	<0.001	<0.001	**supported**
F4	colour augmentation	−0.23	−0.81 to 0.35	0.43	1.00	**not supported**
F5	LEACE erasure	−0.01	−0.10 to 0.07	0.72	1.00	**not supported**
F6	stain normalisation	0.02	−0.41 to 0.44	0.94	1.00	**not supported**
F7	probe-matched adversary	0.000	−0.038 to 0.039	0.99	1.00	**not supported**
F8	pooled dose-response	0.000	−0.005 to 0.005	0.93	1.00	**not supported**

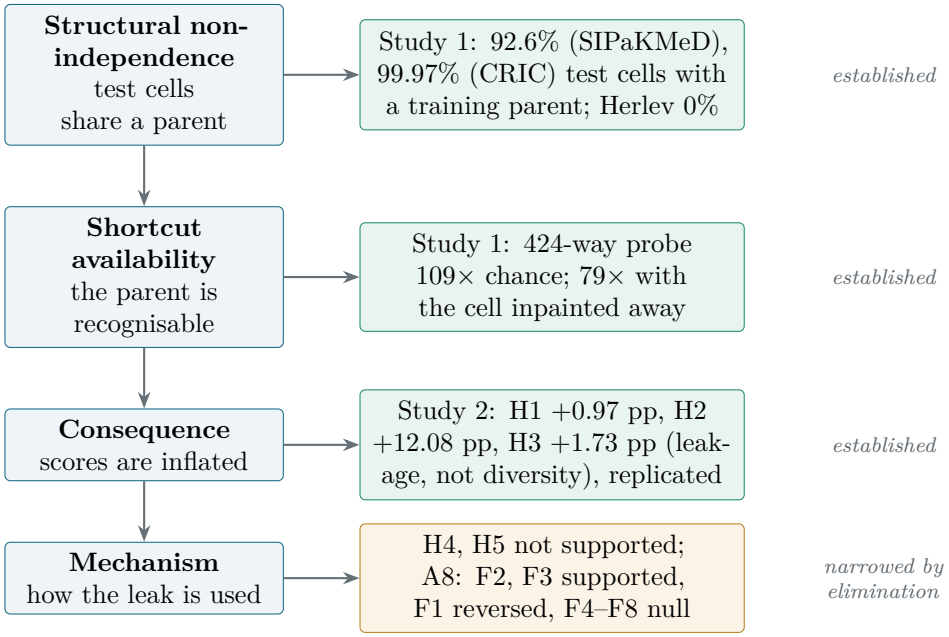


Figure 1.3: The four claims the project separates, and the evidence each received. Study 1 established the first two; Study 2 the third; Amendment 8 narrowed the fourth.

1.4 What we got out of it

Established

- Crop-level splitting inflates accuracy on SIPaKMeD by 0.97 pp (95% CI 0.59–1.39; all twelve architectures positive) and on CRIC by 12.08 pp; the parent-level error rate is 1.34 and 2.27 times the crop-level one.
- The inflation is leakage, not the extra training diversity crop-level splits bring: at matched crop and field counts the leak effect is 1.73 pp and the diversity effect 0.00 pp; the leak effect replicated on fresh seeds (1.47 pp).
- The leak needs no background (F3: 1.86 pp with every background masked) and is tied to the field's colour rendering at test time (F2: stain transfer cuts it from 1.72 to 0.46 pp).
- Neither a linearly decodable field code nor Grad-CAM-visible attention carries it: erasing or adversarially suppressing field identity cut probe recoverability by up to 91% without moving the gap, and Grad-CAM maps of leak and clean models were indistinguishable.

Not supported, reversed, or open

- H4 (a background context route) and H5 (the gap tracks gradient-reversal suppression) were not supported; the ConvNeXt-T adversary was evaded.
- F1 reversed: CRIC models were *not* pulled toward the slide's majority label.
- F4: strong colour augmentation did not prevent leakage, which qualifies F2.
- Parents are not patients; one training recipe was used throughout; Study 2's registrations still need an external timestamp.

The practical message

Probes, saliency maps, background removal, and colour augmentation cannot certify that an evaluation is free of hierarchical leakage. Only a split at the deepest released parent unit removes it, and the closed forms of Chapter 25 predict which benchmarks are exposed before any model is trained.

1.5 Numbers at a glance

Quantity	Value
Trained models (Study 1, Study 2, Amendment 8)	2,606 (325 + 1,341 + 940)
Evaluation-only runs on saved models (Amendment 8)	640
Training epochs logged (Study 2 and Amendment 8)	45,600
GPU-hours, Study 2 / Amendment 8	102.7 / 49.8
Study 2 matrix wall clock	27.4 h (18 Oct 05:07 to 19 Oct 08:28)
Amendment 8 matrix wall clock	12.8 h (19 Oct 14:03 to 20 Oct 02:52)
Failed tasks, retries, diverged runs	0, 0, 0
Smoke tests before registration	30
Commits	82
Registered confirmatory tests	1 primary + 4 (Holm) + 8 (Holm)

Bibliography